

Phase 7

Project Documentation Phase

Technical reference and documentation index

1. Documentation Overview

This document is the documentation index and technical reference for the AI StudyBuddy backend. It summarizes the project, the architecture, the setup and run steps, the API surface and the supporting materials produced in each phase.

2. Project Summary

AI StudyBuddy is an AI-powered learning assistance platform. Its Node.js / Express.js backend follows a RESTful architecture and uses MongoDB with Mongoose, JWT authentication with Role-Based Access Control and bcrypt password hashing, plus the Google Gemini AI API to generate summaries, flashcards, quizzes and personalized study plans from study material uploaded by students.

3. Documentation Set (8 Phases)

#	Phase document	What it contains
1	Brainstorming and Ideation	Problem, pain points, ideas, proposed solution, target users
2	Requirement Analysis	Functional and non-functional requirements, data, software and hardware needs
3	Project Design	Architecture, ER diagram, user flow, MVC, folder structure, API design
4	Project Planning	Epics, tasks, milestones, tools, risks
5	Project Development	Setup commands, environment, core code and schemas
6	Project Testing	Postman test cases, results and screenshots
7	Project Documentation	This reference document
8	Project Demonstration	Demo walkthrough, videos and links

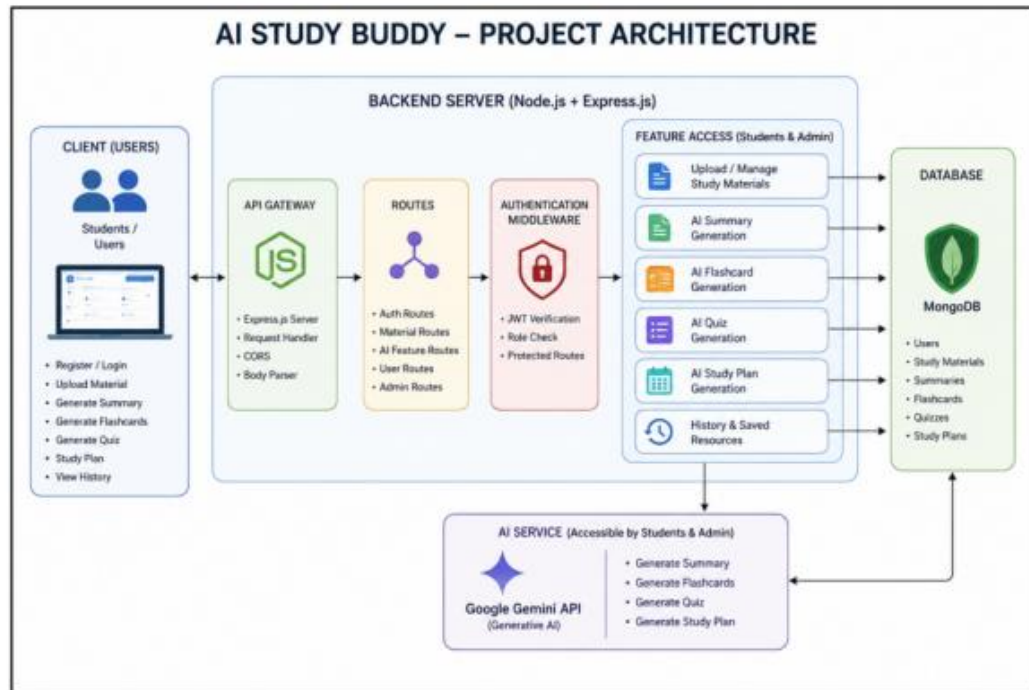
4. Technology Stack

Area	Technology
Runtime	Node.js (v16+), npm (v8+)
Framework	Express.js
Database	MongoDB, Mongoose ODM
Security	JWT (jsonwebtoken), bcryptjs, RBAC
File uploads	Multer (.txt, .md, .pdf, max 5 MB)
AI	Google Gemini 2.5 Flash API
Other	cors, dotenv, nodemon (dev)
Testing	Postman / Thunder Client

5. Key Features

- **Secure user authentication** with JWT and bcrypt-encrypted passwords.
- **Study material management** for uploading and managing AI input.
- **AI summary, flashcard, quiz and study plan generation** using the Google Gemini API.
- **Role-Based Access Control** with different privileges for Students and Administrators.
- **MongoDB data storage** for users, materials and generated resources.

6. Architecture Summary



AI StudyBuddy - Project Architecture

Requests pass through the Express server and JWT middleware to route handlers and controllers, which use Mongoose models for MongoDB and the AI service (Gemini) for content generation. Authentication, business logic, database operations and AI processing are separate modules following the MVC pattern.

7. Setup and Run Guide

- Install Node.js (v16+), npm (v8+), MongoDB and a code editor (Visual Studio Code).
- Clone / open the project and run `npm install`.
- Create `.env` with `PORT`, `MONGO_URI`, the JWT secret and `GEMINI_API_KEY`.
- Make sure MongoDB is running, then start the server with `npm start`.
- Open `http://localhost:5000/`; the health check returns 'AI StudyBuddy API is running'.
- Test the APIs with Postman.

8. Environment Variables

Variable	Purpose
PORT	Port the server listens on (default 5000)
MONGO_URI	MongoDB connection string
JWT secret (JWT_SECRET in .env template; JWT_ACCESS_SECRET in auth middleware)	Secret used to sign and verify JWT access tokens
GEMINI_API_KEY	API key for Google Gemini
CLIENT_URL (optional)	Allowed CORS origin (default http://localhost:3000)

9. API Reference

Method	Endpoint	Description	Auth
POST	/api/auth/register	Register a new user (password hashed)	No
POST	/api/auth/login	Log in and receive JWT tokens	No
POST	/api/materials/upload	Upload study material (form-data: file, title)	Bearer token
POST	/api/materials/:id/summarize	Generate AI summary	Bearer token
POST	/api/materials/:id/flashcards	Generate AI flashcards	Bearer token
POST	/api/materials/:id/quiz	Generate AI multiple-choice quiz	Bearer token
POST	/api/materials/:id/study-plan	Generate AI study plan (goal, hoursPerDay, days)	Bearer token
-	/api/admin/*	Administrator-only user management and monitoring	Bearer token + admin role

The original document's headings list the AI endpoints as /api/ai/flashcards, /api/ai/quiz and /api/ai/study-plan, but the Postman tests use /api/materials/:id/... URLs. The tested URLs are used above.

10. Roles and Responsibilities

Role	Access
Student	Register, log in, upload materials, use AI features, view and manage own generated resources
Administrator	Manage users, monitor activity and AI usage, review logs, system-level tasks

11. Supporting Materials

- Backend Development Demo: AI_Studybuddy_Backend_Demo.mkv
- API Testing / Execution Demo: AI_Studybuddy_Execution_Demo.mkv
- Demo video link:
https://drive.google.com/file/d/11ShkBRp-qSXogvOqFgn4lnZa3CQok1GFr/view?usp=drive_link
- Source code: shared through the project's Code drive link